HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and configured **Python scripts** to automate repetitive engineering tasks, reducing manual effort by approximately **40%** for data extraction and reporting workflows.
- Designed and deployed **Java Spring Boot REST APIs** for cross-service communication, implementing robust error handling and request validation to support enterprise system integrations.
- Collaborated with **discipline leads** to map current processes, identify pain points, and define clear digital requirements for automation opportunities within Life Science engineering projects.
- Managed solution initiatives through definition, development, testing, and pilot phases, meticulously recording assumptions, risks, and lessons learned for **iterative improvement**.
- Supported the evaluation and initial implementation of **AI capabilities**, focusing on structured data extraction from project documents and natural-language interfaces.
- Built and coordinated **prototypes and proofs of concept** using Microsoft 365, Autodesk Platform Services, and data automation technologies to demonstrate digital engineering solutions.
- Containerized **Java applications** with Docker for consistent deployment environments and managed basic configurations within an AWS EC2 setup.

SKILLS

Engineering & Digital Platforms – Autodesk Revit, Plant 3D, AutoCAD, Navisworks, Autodesk Forma, Microsoft 365, Power Platform, Dataverse, SharePoint, Power BI, Autodesk Platform Services

Programming & Scripting – Python, Java, SQL, Spring Boot, REST APIs

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Git, Jenkins, CI/CD

Data & AI – Data-driven reporting, AI-assisted engineering, structured data extraction, anomaly detection, knowledge retrieval

Project Management & Support – Change management, user adoption, training, technical support, process mapping, requirements gathering

PROJECTS

Digital Workflow Automation for Engineering Design

- Developed a **Python-based workflow** to automate repetitive design checks and data extraction from engineering models (e.g., Revit, AutoCAD), reducing task completion time by an estimated **30%**.
- Integrated the automation script with **Autodesk Platform Services** to extract relevant data points, creating structured outputs for downstream analysis and reporting.
- Documented the workflow, prepared user guidance, and conducted initial demonstrations for engineering teams to support adoption and gather feedback for refinement.

Life Science Project Data Integration Hub

- Engineered a **Java Spring Boot service** to aggregate data from various project sources including drawings, schedules, and reports, creating a unified data view.
- Implemented **SQL queries** against PostgreSQL to validate data integrity and structure information for use in **Power BI dashboards** supporting project decision-making.
- Managed the deployment of the service using Docker and basic **AWS EC2** configurations, ensuring reliable data flow for project reporting.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)